2311018

” ” Word Embedding

1.
2.
3. Logistic Regression
4. Word2Vec GloVe
5. BERT Fine-tuning
6. Accuracy Macro-F1
7.

2.1

/		
	Windows 10/11 (x64)	
Python	3.12.7	
scikit-learn	1.9.0	
gensim	4.3.x	Word2Vec
transformers	5.x	BERT

2.2

- **NYT** 11,519 3 sports/politics/business
- **AG News** 90,000 Word2Vec

2.3

- 80% 9,215
- 20% 2,304
- 42

3.1 Accuracy

3.2 Macro-F1

F1 Score F1 Macro-F1

Task 1 Bag-of-Words 30

4.1 Binary Bag of Words

- x = 1 x = 0

4.2 Word Frequency

- x = c

4.3

	Accuracy	Macro-F1
Binary BoW	0.9848	0.9634
Word Frequency	**0.9870**	**0.9687**

Task 2 Word Embedding 50

$$e(d) = (1/n) \times \Sigma \quad e(w)$$

5.1 Pre-trained GloVe glove.6B.100d

- 400,000
- 100

5.2 Word2Vec Trained on AG News

- Gensim Word2Vec
- 100 5 10

5.3 Word2Vec Trained on NYT

- 5.2

5.4

	Accuracy	Macro-F1
GloVe (Pre-trained)	**0.9813**	**0.9573**
Word2Vec (AG News)	0.9774	0.9492
Word2Vec (NYT)	0.9783	0.9484

Task 3 BERT 20

6.1

- google-bert/bert-base-uncased
- max_length = 64
- 3 epochs

6.2

	Accuracy	Macro-F1
BERT (Fine-tuned)	0.9766	0.9469

6.3

BERT max_length=64 BERT 8%

7.1

		Accuracy	Macro-F1
Task 1	Word Frequency	**0.9870**	**0.9687**
Task 2	GloVe (Pre-trained)	0.9813	0.9573
Task 3	BERT (Fine-tuned)	0.9766	0.9469

7.2

1.
2. **Macro-F1 Accuracy**
3. “ ” ” ”
4. BERT 105 BoW 2

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1. ” ” ” ” BERT ” ”
2. ” ” BERT ” ” ” ”
3. batch size

9.1

```
HW-1/
  homework1.py              # Task 1:
  homework1_word2vec.py     # Task 2.2 & 2.3: Word2Vec
  homework1_glove.py        # Task 2.1: GloVe
  homework1_bert.py         # Task 3: BERT
  nyt.csv / ag.csv          #
  task*_results.csv         #
```

9.2

```
pip install numpy pandas scikit-learn gensim transformers torch accelerate
python homework1.py          # Task 1
python homework1_glove.py    # Task 2.1
python homework1_word2vec.py # Task 2.2 & 2.3
python homework1_bert.py     # Task 3
```